

Algebra 1 - Review

1. Write an equation in point-slope form of the line that passes through the point $(-8, 3)$ and has a slope of $\frac{1}{4}$.

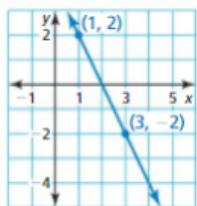
$$3 = \frac{1}{4}(-8) + b$$

$$3 = -2 + b$$

$$b = 5$$

$$y = \frac{1}{4}x + 5$$

2. Write an equation in slope-intercept form of the line shown.



~~$y = -2x + 3$~~

$$y = -2x + 4$$

$$-2 = -2(3) + b$$

$$-2 = -6 + b$$

$$4 = b$$

3. Write a linear function f with the values $f(4) = -2$ and $f(8) = 4$.

$$f(x) = \frac{3}{2}x - 8$$

$$-2 = \frac{3}{2}(4) + b \quad (4, -2) \quad (8, 4)$$

$$-2 = 6 + b$$

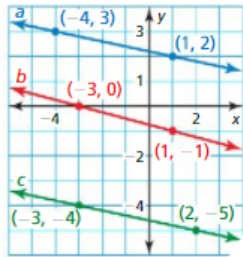
$$b = -8$$

$$\frac{6}{4} = \frac{3}{2}$$

~~2(4)~~

$$+4 \left(\begin{matrix} (4, -2) \\ (8, 4) \end{matrix} \right) + 6$$

4. Determine which of the lines are parallel.



Lines A and C

5.

Write an equation of the line that passes through $(5, -4)$ and is parallel to the line $y = 2x + 3$.

$$y = 2x + b$$

$$-4 = 2(5) + b$$

$$b = -14$$

$$\underline{y = 2x - 14}$$

6.

Determine which of the lines, if any, are parallel or perpendicular.

Line a: $y = 4x + 2$

Line b: $x + 4y = 3$

Line c: $-8y - 2x = 16$

$$y = 4x + 2 \quad 4y = -x + 3$$

$$y = -\frac{1}{4}x + \frac{3}{4}$$

$$\cancel{8y} = 2x + 16$$

$$y = -\frac{1}{4}x - 2$$

(Lines B and C) Parallel

Lines A & B
Lines A & C

Perpendicular